Atanas Stefanov

Curriculum vitae

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Education

09/2019 – 06/2023 **MSci Astrophysics**, *University College London*, London, UK, 80.79% (First Class Honours)
EQF7 Main modules: Physics of Stars, Physics of Exoplanets, Astronomical Spectroscopy, Quantum Physics. Mathematical Methods, Theory of Dynamical Systems, Techniques of High-Performance Computing.

09/2014 – 06/2019 **System Programming**, *MHS "Acad. Kiril Popov"*, Plovdiv, BG, 5.82 (equiv. A*A)
EQF4 A professional qualification obtained alongside secondary education.
Main modules: Electronics, Object-Oriented Programming, Web Development, Database Management.

09/2014 – 05/2019 **Secondary Education**, *MHS "Acad. Kiril Popov"*, Plovdiv, BG, 5.97 (equiv. A*A*)
EQF4 Main modules: Mathematics, Natural Sciences, Bulgarian, English, Russian.

Experience

12/2022 – 03/2023 **Research & Development Associate**, *Synaptic*, Sofia, BG
Technical documentation and consultancy work in the field of natural sciences and information technologies. Engaged with the data visualisation and documentation of a prototype device in the area of rehabilitation and well-being, using Python and LaTeX.

06/2021 – 08/2021 **Research Intern**, *University College London*, London, UK
A Brian Duff studentship award by the UCL Astrophysics group, with topic "Exploration of exoplanet transits around gravity-darkened stars". Documented the use of TESS data products and the Python packages Lightkurve and eleanor. Worked with exoBush, a Fortran-based spectroscopic and photometric simulation tool described in Howarth & Smith (2001). Developed PEPPER, a Python wrapper that: (1) rapidly executes exoBush across a parameter space, (2) provides low-level tools for working with exoBush inputs/outputs, (3) provides affine-invariant Markov chain Monte Carlo fitting through emcee.

10/2018 – 03/2019
10/2017 – 03/2018 **Web Development Intern**, *Viscomp Ltd.*, Plovdiv, BG
Two part-time internships, each over a span of six months. Collaborated on a prototype web-based grade-book for a local university. Worked primarily on back-end development using PHP and MySQL.

Projects

04/2018 – 09/2018 **Colour Relations in Young Stellar Objects**, EUCYS 2018, Dublin, Ireland
Analysed the possibility of finding T Tauri stars through photometric selection. Represented Bulgaria in the European Union Contest for Young Scientists (EUCYS) and won the special JRC Award.
`eucys2018.com/project_details?id=121`

Programming and markup

Currently using

LaTeX	*6 years*
Python	*4 years*
Wolfram	*2 years*
Go	*2 years*

Prior experience

C++	*4 years*
PHP	*4 years*
SQL	*4 years*
CSS	*2 years*
JS	*2 years*

Research work

S. Y. Stefanov and A. K. Stefanov. Tilted discs in six poorly studied cataclysmic variables. *Monthly Notices of the Royal Astronomical Society*, 520(3):3355–3367, April 2023.

S. Y. Stefanov, A. Kurtenkov, and A. K. Stefanov. Spectroscopy of probable dwarf nova AT 2022piu. *The Astronomer's Telegram*, 15538:1, August 2022.

S. Y. Stefanov, A. K. Stefanov, and A. Kurtenkov. Transient Classification Report for 2022-08-01. *Transient Name Server Classification Report*, 2022-2168:1, August 2022.